MIEB: A Massive Benchmark for Image and Image-Text Embeddings

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Abstract

Image and image-text embedding models are now used for retrieval, clustering, classification, document understanding, and multimodal question answering, but evaluation remains fragmented across narrow, task-specific benchmarks. This makes it difficult to understand whether a strong model on one setting transfers to others, especially for multilingual retrieval, visual text understanding, interleaved inputs, and compositional reasoning. We introduce MIEB, a unified benchmark for frozen image and image-text embeddings built in the MTEB framework. MIEB covers 130 tasks from 8 categories, spans 38 languages, and standardizes evaluation across retrieval, document understanding, linear probing, zero-shot classification, clustering, compositionality, vision-centric QA, and Visual STS. We also define MIEB-lite, a 51-task subset designed to retain the main diagnostic value of the full benchmark at substantially lower cost. Experiments on representative encoder families and MLLM-derived embedding models show that no single model dominates all categories: contrastive encoders remain strongest on linear probing, zero-shot classification, and clustering, while MLLM-based embeddings are stronger on multilingual retrieval, document understanding, and visual text similarity. The best overall score on the full benchmark is 62.5, indicating considerable headroom. MIEB provides a reproducible, public evaluation protocol that exposes complementary strengths and weaknesses that narrower benchmarks miss.

1. Introduction

Embedding models have become a core interface for multimodal systems. The same representation may be used to retrieve images from text, retrieve captions from screenshots, cluster image collections, support few-shot classifiers, or serve as the vision encoder inside a larger multimodal system. Yet evaluation practice remains fragmented. Models are often reported on a small number of retrieval datasets, a single zero-shot classification suite, or a document benchmark in isolation. As a result, it is still unclear how well current image and image-text embeddings generalize across task formats, modalities, and languages.

This fragmentation matters for both science and deployment. From a scientific perspective, it obscures which capabilities transfer and which remain specialized. From a systems perspective, it makes model selection brittle: a model chosen for image-text retrieval may underperform on document understanding, multilingual search, or compositional reasoning. Text embedding research addressed a similar problem with MTEB and its multilingual extensions, which standardized evaluation across many tasks and made cross-task trade-offs visible [5, 15]. A comparable benchmark for image and image-text embeddings has been missing.

We introduce MIEB, the Massive Image Embedding Benchmark, to fill this gap. MIEB unifies 130 public tasks into a single evaluation protocol implemented in the MTEB codebase. The benchmark covers 8 categories: retrieval, document understanding, linear probing, zero-shot classification, clustering, compositionality, vision-centric QA, and Visual STS. It spans image-only, text-to-image, image-to-text, and interleaved settings, and extends evaluation beyond English to 38 languages. The design goal is not to propose a new embedding model, but to provide a benchmark that more reliably measures how frozen representations transfer across realistic multimodal use cases.

Our empirical results support the need for a broad benchmark. The best overall model on MIEB is Voyage-multimodal-3 with a mean score of 62.5, followed by E5-V at 58.2 and SigLIP So400m at 57.1. However, these aggregate numbers hide sharp specialization. Contrastive encoders such as SigLIP, CLIP, and EVA-CLIP are strongest on zero-shot classification, linear probing, and clustering. MLLM-based embeddings perform better on multilingual retrieval, document understanding, and Visual STS, with E5-V reaching 74.3 average on multilingual retrieval and Voyage-multimodal-3 reaching 71.1 on document understanding and 81.8 on English Visual STS. Compositionality remains challenging across families, with the best category score only 46.3.

Our contributions are threefold. First, we present a large-scale, unified benchmark for image and image-text embeddings with public code and leaderboard support. Second, we introduce Visual STS as a diagnostic protocol for visual text understanding by rendering semantic textual similarity pairs as images. Third, we provide MIEB-lite, a smaller subset that preserves the same overall winner and a simi-

lar ranking pattern while substantially reducing evaluation cost.

2. Related Work

Embedding benchmarks. The closest conceptual predecessors to MIEB are MTEB and MMTEB, which standardized evaluation for text embeddings across diverse tasks and languages [5, 15]. Their main lesson is that single-dataset evaluation overestimates universality: models that excel on one task family often lag on others. MIEB adopts this benchmark philosophy for multimodal embeddings and extends it to image-only, image-text, interleaved, and multilingual settings.

Multimodal retrieval and document benchmarks. Prior multimodal evaluation has mostly focused on retrieval or a narrow application domain. CLIP established strong zero-shot transfer but did not provide a comprehensive cross-task benchmark [16]. Subsequent models such as SigLIP, Jina-CLIP, E5-V, and VLM2Vec improved particular aspects of multimodal retrieval or general-purpose embeddings [9, 10, 12, 23]. Benchmarks such as UniIR/M-BEIR and ViDoRe broaden evaluation for multimodal retrieval and document retrieval respectively, but they remain centered on retrieval-style tasks [6, 20]. MIEB differs in that retrieval is only one component of a broader diagnostic suite that also includes clustering, classification, compositionality, vision-centric QA, and visual similarity.

Diagnostic vision-language benchmarks. Several datasets probe specific multimodal weaknesses, including compositionality and visual perception. ARO, SugarCrepe, and Winoground test attribute, relation, and order sensitivity [8, 18, 22]; Blink targets low-level visual perception [7]; and InfoSeek measures information-seeking vision-language reasoning [2]. These resources are valuable but intentionally narrow. MIEB incorporates such datasets as benchmark components rather than alternatives, allowing us to measure how diagnostic tasks relate to more standard retrieval and classification settings.

3. Benchmark Design

3.1. Task Coverage

MIEB aggregates 130 tasks across 8 categories, summarized in Figure 1 and Table 1. The benchmark includes 125 English tasks and extends to 130 tasks in the multilingual setting through multilingual retrieval and Visual STS. Coverage spans image-only, text-to-image, image-to-text, image-text-to-image, and other interleaved retrieval formats, as well as both discriminative and similarity-based evaluation.

The benchmark design follows four principles. First, *breadth*: MIEB should cover the dominant uses of frozen multimodal embeddings rather than a single downstream task. Second, *diagnostic value*: the benchmark should contain both standard workloads and targeted stress tests such as compositionality and visual text understanding. Third, *multilinguality*: cross-lingual and multilingual retrieval should be part of the main benchmark rather than an afterthought. Fourth, *reproducibility*: all tasks are integrated into the MTEB framework with explicit dataset versions and shared evaluation code.

3.2. Task Sources

MIEB is built entirely from public datasets and reuses existing human annotations. The retrieval suite includes standard image-text retrieval, image-image retrieval, interleaved retrieval, and document retrieval. Examples include Flickr30k- and COCO-style retrieval, multilingual retrieval from xFlickr&CO, XM3600, and WIT, document retrieval tasks from the ViDoRe family, entity-rich and information-seeking tasks such as EDIS and InfoSeek, and compositional retrieval from CIRR and FashionIQ [2, 6, 14, 17]. The compositionality suite combines ARO, SugarCrepe, Winoground, and ImageCoDe [8, 13, 18, 22]. Vision-centric QA includes Blink and CV-Bench-style tasks [7, 19]. Linear probing and zero-shot classification cover standard vision datasets such as ImageNet-1k, CIFAR, Oxford Pets, Oxford Flowers, Food101, and Stanford Cars [4, 11].

3.3. Visual STS

An important under-evaluated capability of image encoders is visual text understanding: whether a model can represent semantic similarity when the textual input is presented as pixels rather than tokens. To test this, MIEB includes Visual STS, which renders sentence pairs from STS datasets as images and evaluates whether semantically similar rendered sentences remain close in the embedding space. The resulting tasks cover English and multilingual settings and are implemented as image-image similarity tasks scored by Spearman correlation. This protocol is closely connected to recent work on visually grounded sentence representation learning [21], but here it is used as a benchmark component rather than as a training method.

3.4. Evaluation Protocol

MIEB evaluates frozen embeddings. Most tasks are zero-shot: the model produces embeddings directly and is scored with the task's main metric. Retrieval and document understanding use ranking metrics, predominantly nDCG-based task scores. Clustering uses K-means-style clustering with normalized mutual information (NMI). Zero-shot classification, compositionality, and vision-centric QA use accuracy. Visual STS uses Spearman correlation.

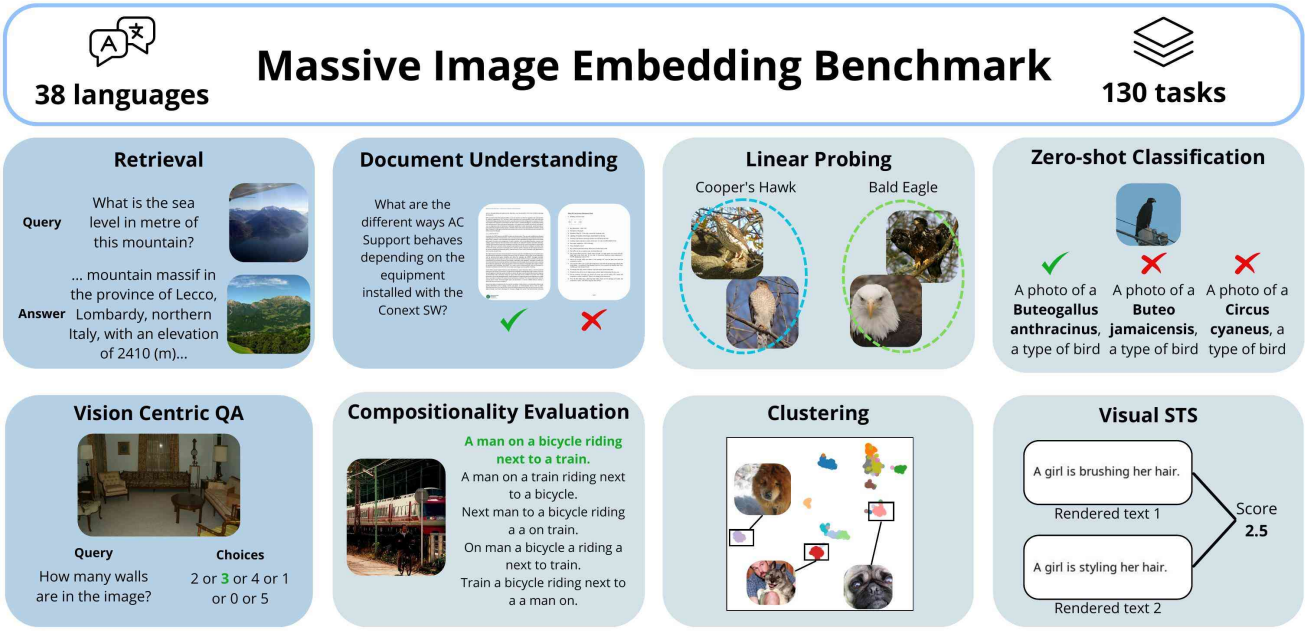

Figure 1. **Overview of MIEB.** The benchmark covers eight evaluation categories with image-only, cross-modal, multilingual, and interleaved settings. Rather than optimizing for a single task family, MIEB is designed to reveal transfer and specialization across diverse embedding use cases.

Task category	Example abilities assessed	# Tasks	# Languages	Modalities
Retrieval	cross-modal/-lingual matching	45	38	i-i; i-t; t-i; it-i; it-t; i-it; t-it; it-it; i-t
Document Understanding (Retrieval)	OCR abilities	10	2	t-i; i-t; it-t
Linear Probing (Classification)	information encoded	22	1	i-i; i-i
Clustering	embedding space consistency	5	1	i-i
Zero-shot Classification	cross-modal matching	23	1	i-t; i-t
Compositionality Evaluation (PairClassification)	reasoning with confounders	7	1	i-t; t-i
Vision-centric QA (Retrieval)	counting, object detection	6	1	it-t; it-i
Visual STS	OCR abilities	9	12	i-i
MIEB	all	130	38	all
MIEB-lite	all	51	38	all

Table 1. **An overview of MIEB tasks.** In brackets behind task categories, we denote the task type implementation in the code, e.g., our document understanding tasks use our retrieval implementation. We denote the modalities involved in both sides of the evaluation (e.g., queries and documents in retrieval; images and labels in zero-shot classification) with i=image, t=text.

Linear probing is the only setting with supervised adaptation. For the image classification tasks, MIEB trains a logistic regression classifier on top of frozen embeddings with 16 labeled examples per class and a maximum of 100 optimization iterations, following the task definitions implemented in the codebase. This gives a controlled measure of how much class information is linearly recoverable without fine-tuning the encoder itself.

4. Benchmark Validation

4.1. Coverage and Diversity

MIEB's first validation goal is scope: does the benchmark cover genuinely different evaluation settings rather than many near-duplicates? Table 1 shows diversity along three axes. The first is task family: MIEB includes retrieval, similarity, clustering, multiple-choice reasoning, and both zero-shot and few-shot classification. The second is modality: the benchmark mixes image-only, cross-modal, and interleaved settings. The third is language: multilingual retrieval tasks span 38 languages, and multilingual Visual STS covers a distinct subset of languages focused on visually ren-

Model Name	# Params (M)	Runtime (NVIDIA H100 GPU hours)		
		MIEB	MIEB-lite	Reduction %
E5-V	8360	264.0	46.4	82.4% ↓
CLIP (base-patch32)	151	16.6	4.5	72.9% ↓

Table 2. **MIEB vs. MIEB-lite runtime comparison.**

dered text similarity.

4.2. Quality and Reproducibility

MIEB does not introduce new manual annotations. Instead, it standardizes public datasets with existing human labels, captions, or similarity judgments. This choice improves reproducibility and makes benchmark maintenance practical, but it also means MIEB inherits source-dataset artifacts and annotation biases. To mitigate benchmark-level inconsistency, all tasks are versioned and evaluated through a common implementation framework. That standardization is especially important because MIEB mixes historically separate protocols such as retrieval, classification, STS, and document understanding.

4.3. MIEB-lite

The full benchmark is intentionally broad, but broad evaluation is expensive. We therefore include MIEB-lite, a 51-task subset selected through the benchmark's task-selection pipeline to preserve the main ranking signal at lower cost. The subset still covers all major categories and languages present in the full benchmark. Importantly, it retains the same overall winner as the full benchmark: Voyage-multimodal-3 ranks first on both MIEB and MIEB-lite in Table 4.

Runtime reductions are substantial. As shown in Table 2, MIEB-lite reduces evaluation time from 264.0 to 46.4 H100 GPU hours for E5-V, an 82.4% reduction, and from 16.6 to 4.5 hours for CLIP ViT-B/32, a 72.9% reduction. It also remains highly rank-preserving, with Spearman 0.992 against the full benchmark. This makes broad multimodal evaluation more practical without collapsing the benchmark into a single narrow task family.

4.4. Sanity Check for Visual STS

Visual STS should be related to, but not identical to, conventional text STS. Table 3 provides a concrete sanity check using E5-V. The same model scores 86.0 on standard STS but 79.3 on Visual STS. The gap is informative: rendering text as images preserves enough semantic structure for a strong model to transfer, but the benchmark remains meaningfully harder than token-based STS.

5. Experimental Setup

Models. We evaluate a representative mix of contrastive encoders and MLLM-derived embedding models. The model pool includes CLIP and OpenCLIP variants [3, 16], SigLIP variants [23], EVA-CLIP, E5-V [9], VLM2Vec [10], and Voyage-multimodal-3 [1]. This choice is deliberate: the benchmark is intended to compare model families with different training objectives and inductive biases rather than to optimize a single leaderboard entry.

	12	13	14	15	16	17	b	avg.
STS*	80.0	89.9	85.7	89.1	85.9	87.9	83.5	86.0
v-STS (ours)	73.2	78.2	74.9	84.2	79.5	85.8	79.4	79.3

Table 3. **E5-V performance on regular STS and our Visual STS.** *: numbers from Jiang et al. [9]. Columns are STS12-17 and STS-b.

Implementation. All evaluations are run through the public MTEB codebase [5, 15]. This provides a unified task interface, standard result serialization, and shared leaderboard infrastructure. For image-only encoders, we evaluate the image-compatible subset of tasks; for multimodal models, we evaluate the full suite, including text-conditioned and interleaved tasks.

Reporting. We report per-category averages as well as aggregate means for the English-only and multilingual benchmarks. The aggregate mean should be interpreted as a compact summary, not as a claim that all categories are equally important or equally difficult. The per-category breakdown is therefore central to the analysis.

6. Benchmark Results and Analysis

6.1. Overall Results

The headline result is that no single model dominates MIEB across all categories. Voyage-multimodal-3 achieves the best overall multilingual mean at 62.5 and also ranks first on MIEB-lite at 58.1. E5-V is second overall at 58.2, and SigLIP So400m is the strongest pure encoder at 57.1. However, these overall scores mask sharp specialization.

Contrastive encoders are strongest on structure-preserving visual tasks. EVA02-CLIP-bigE-14-plus achieves the best clustering score (92.4), the best linear probing score (86.0), and ties for the best zero-shot classification score on the full benchmark. SigLIP variants are consistently strong on zero-shot classification and linear probing, and CLIP ViT-bigG achieves the best retrieval category score (41.5). In contrast, MLLM-derived embeddings are strongest where language and visual text matter most: Voyage-multimodal-3 leads document understanding (71.1) and English Visual STS (81.8), while E5-V achieves the strongest multilingual retrieval average among the top families.

MIEB Full (130 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (45)**	**Clus. (5)**	**ZS. (23)**	**LP. (22)**	**Cmp. (7)**	**VC. (6)**	**Doc. (10)**	**vSTS (en) (7)**	**Rtrv. (m) (3 (55))**	**vSTS (x&m) (2 (19))**	**Mean (en) (125)**	**Mean (m) (130)**
Voyage-multimodal-3	MLLM	38.8	82.4	58.2	71.3	43.5	48.6	**71.1**	**81.8**	58.9	**70.4**	**62.0**	**62.5**
E5-V	MLLM	34.0	70.0	50.0	74.5	**46.3**	51.9	62.7	79.3	**66.6**	46.3	58.6	58.2
siglip-so400m-patch14-384	Enc.	40.8	82.1	**70.8**	84.6	40.4	46.3	56.4	68.0	40.2	41.4	61.2	57.1
siglip-large-patch16-384	Enc.	39.9	79.9	68.0	83.7	39.7	45.4	53.3	69.5	51.1	39.8	59.9	57.0
siglip-large-patch16-256	Enc.	38.8	82.1	67.7	82.5	40.8	44.9	39.4	67.4	49.8	38.1	57.9	55.2
siglip-base-patch16-512	Enc.	38.1	74.7	64.1	80.9	37.5	53.2	52.1	67.7	43.2	38.1	58.5	54.9
CLIP-ViT-bigG-14-laion2B	Enc.	**41.5**	85.6	69.4	83.6	42.4	43.2	43.2	70.9	28.0	34.5	60.0	54.2
siglip-base-patch16-384	Enc.	37.7	76.3	64.1	80.6	38.5	52.8	45.0	67.0	42.5	37.5	57.8	54.2
EVA02-CLIP-bigE-14-plus	Enc.	40.1	**92.4**	70.8	**86.0**	45.7	39.4	32.3	72.0	27.8	28.2	59.8	53.5
CLIP-ViT-L-14-DataComp.XL	Enc.	38.1	86.4	68.4	82.0	39.1	52.3	38.6	69.9	23.8	35.8	59.4	53.4
siglip-base-patch16-256(m)	Enc.	35.6	74.6	61.2	78.9	38.1	51.3	26.4	65.5	59.2	40.3	53.9	53.1
CLIP-ViT-H-14-laion2B	Enc.	39.7	83.9	67.5	82.5	42.0	45.8	40.4	65.5	25.5	33.9	58.4	52.7
CLIP-ViT-g-14-laion2B	Enc.	39.8	82.7	67.9	82.8	41.9	44.2	37.6	69.1	25.9	31.7	58.3	52.4
EVA02-CLIP-bigE-14	Enc.	39.0	89.4	69.3	84.5	42.4	43.6	31.6	68.8	25.5	28.3	58.6	52.2
siglip-base-patch16-256	Enc.	36.6	75.2	63.1	79.7	39.5	52.2	31.7	66.2	41.3	34.4	55.5	52.0
siglip-base-patch16-224	Enc.	36.3	74.5	62.6	79.3	39.8	51.1	26.2	64.3	41.2	33.5	54.3	50.9
CLIP-ViT-L-14-laion2B	Enc.	38.0	83.5	65.8	81.2	40.8	45.9	36.3	65.8	23.0	26.0	57.2	50.6
VLM2Vec-LoRA	MLLM	27.7	72.6	46.3	62.0	34.6	62.0	49.7	72.6	34.9	42.2	53.4	50.5
VLM2Vec-Full	MLLM	27.6	70.7	46.3	62.0	35.4	**62.1**	49.8	72.6	35.0	42.2	53.3	50.4
clip-vit-large-patch14	Enc.	33.7	76.4	62.1	80.1	44.8	44.1	38.0	64.5	20.2	35.1	55.4	49.9

MIEB-lite (51 tasks)													
Model Name (↓)	**Model Type**	**Rtrv. (11)**	**Clus. (2)**	**ZS. (7)**	**LP. (8)**	**Cmp. (6)**	**VC. (5)**	**Doc. (6)**	**vSTS (en) (2)**	**Rtrv. (m) (2 (47))**	**vSTS (x&m) (2 (19))**	**Mean (en) (47)**	**Mean (m) (51)**
Voyage-multimodal-3	MLLM	33.2	76.6	48.6	69.3	35.8	50.0	**63.5**	**84.2**	49.0	**70.4**	**57.7**	**58.1**
siglip-so400m-patch14-384	Enc.	32.4	75.9	73.8	78.8	32.8	48.0	46.9	69.6	35.4	41.4	57.3	53.5
siglip-large-patch16-384	Enc.	31.9	75.2	71.3	77.7	32.1	46.8	44.9	69.6	43.5	39.8	56.2	53.3
E5-V	MLLM	26.9	51.7	36.2	70.6	**39.4**	52.6	56.0	81.2	**58.3**	46.3	51.8	51.9
siglip-large-patch16-256	Enc.	31.0	76.5	70.3	76.3	33.4	46.5	31.9	67.6	42.6	38.1	54.2	51.4
CLIP-ViT-bigG-14-laion2B	Enc.	34.2	80.8	72.4	77.8	35.0	43.0	35.5	73.4	26.2	34.5	56.5	51.3
siglip-base-patch16-512	Enc.	30.8	69.7	66.3	74.6	29.7	55.5	42.6	67.1	34.8	38.1	54.5	50.9
EVA02-CLIP-bigE-14-plus	Enc.	**35.2**	**87.3**	**74.0**	**80.0**	38.9	38.8	26.2	73.7	26.0	28.2	56.8	50.8
siglip-base-patch16-384	Enc.	30.6	72.2	66.0	74.4	31.0	55.1	37.1	66.9	34.5	37.5	54.1	50.5
CLIP-ViT-L-14-DataComp.XL	Enc.	31.0	80.4	69.4	75.3	31.6	54.9	30.8	72.5	22.6	35.8	55.7	50.4
CLIP-ViT-H-14-laion2B	Enc.	32.8	79.3	69.4	76.8	34.8	46.8	33.7	68.3	23.9	33.9	55.2	50.0
EVA02-CLIP-bigE-14	Enc.	34.3	86.7	73.0	78.3	35.1	44.4	25.1	69.9	23.9	28.3	55.9	49.9
siglip-base-patch16-256(m)	Enc.	28.2	68.2	63.2	73.4	30.7	53.3	22.9	63.7	52.9	40.3	50.4	49.7
CLIP-ViT-g-14-laion2B	Enc.	33.5	76.8	69.6	77.3	34.7	45.0	29.9	71.6	24.2	31.7	54.8	49.4
siglip-base-patch16-256	Enc.	29.5	69.6	65.6	73.6	32.2	54.4	25.0	66.1	33.5	34.4	52.0	48.4
CLIP-ViT-L-14-laion2B	Enc.	31.1	76.4	67.8	75.9	33.6	46.9	28.7	68.7	21.4	26.0	53.6	47.6
clip-vit-large-patch14	Enc.	26.7	71.3	63.8	74.5	**39.4**	44.9	29.4	69.4	19.8	35.1	52.4	47.4
siglip-base-patch16-224	Enc.	29.3	68.4	65.0	73.5	32.5	53.0	20.9	64.2	33.6	33.5	50.8	47.4
CLIP-ViT-B-16-DataComp.XL	Enc.	28.3	73.6	61.9	73.2	31.4	56.9	22.7	69.7	19.9	28.5	52.2	46.6
VLM2Vec-LoRA	MLLM	21.0	66.3	32.1	64.8	29.4	**65.3**	42.7	70.9	24.8	42.2	49.1	46.0

Table 4. **MIEB results broken down by task categories for the top 20 models.** We provide averages of both English and multilingual tasks. Models are ranked by the Mean (m) column. Shortcuts are x=Crosslingual, m=Multilingual, en=English, and task categories from Figure 1. We refer to the leaderboard for the latest version: https://hf.co/spaces/mteb/leaderboard

The benchmark is far from saturated. Even the best overall mean is 62.5, and several diagnostic categories remain difficult. Compositionality is the clearest example: the best score is only 46.3, achieved by E5-V, indicating that strong generic retrieval or classification performance does not imply robust handling of attribute binding, relation understanding, or order sensitivity.

6.2. Multilingual Retrieval

Multilingual retrieval is one of the clearest places where the benchmark changes the model ranking. Table 5 shows that E5-V reaches 74.3 average across xFlickr&CO, XM3600, and WIT, clearly outperforming the strongest SigLIP family representative at 66.8. It also shows much lower cross-language variance than older CLIP-style baselines. This matters because it suggests that multilingual transfer is not a simple byproduct of strong English retrieval; it is a distinct capability that MIEB exposes directly.

6.3. Visual Text Understanding

Visual STS provides a complementary diagnostic for visually rendered text. Table 3 shows that E5-V transfers

Model Name	xFlickr&CO		XM3600		WIT		avg.	
	avg.	var.	avg.	var.	avg.	var.	avg.	var.
E5-V	**90.8**	**0.1**	**74.8**	3.5	**57.3**	0.6	**74.3**	1.4
SigLIP	80.4	1.2	65.6	5.3	54.4	1.3	66.8	2.6
VISTA (m3)	65.3	0.2	48.5	**2.0**	49.3	**0.4**	54.4	**0.9**
VLM2Vec	63.8	3.8	27.0	4.7	31.7	2.5	40.8	3.6
Open-CLIP	35.9	9.3	20.5	6.0	37.8	6.5	31.4	7.3
EVA02-CLIP	35.6	9.4	20.1	6.0	37.4	6.4	31.0	7.2

Table 5. **Performance of models on multilingual retrieval tasks across 38 languages.** We compute the average performance across languages (avg) and the respective variance (var). We take the best variant from each top-6 model family.

strongly from token-based STS to rendered-text STS, but not perfectly: the 6.7-point gap between 86.0 on STS and 79.3 on Visual STS shows that OCR-like semantic understanding remains a real challenge. In the full benchmark, Voyage-multimodal-3 leads English Visual STS, while E5-V remains especially competitive on multilingual retrieval and compositionality. Together, these results suggest that visually grounded language understanding is a major differentiator among modern multimodal embeddings.

6.4. What MIEB Reveals About Current Models

Three patterns stand out. First, broad generality remains elusive: models that are strongest on classification are not necessarily strongest on multilingual retrieval or document tasks. Second, MLLM-derived embeddings appear to offer a real advantage on OCR-heavy and language-rich settings, while classic contrastive encoders still excel on fine-grained visual discrimination. Third, benchmark choice changes conclusions. A retrieval-only benchmark would favor a different set of models than a classification-heavy benchmark, whereas MIEB makes these trade-offs explicit in one place.

6.5. Additional Diagnostics

MIEB is also useful beyond averages. Figure 2 gives qualitative slices for clustering and linear probing, and Figure 3 shows that Visual STS tracks OCR-heavy MLLM benchmarks far more closely than 3D-heavy perception benchmarks [19].

7. Discussion

MIEB is intended as an evaluation instrument, not a claim that a single score can summarize all aspects of multimodal intelligence. It measures the quality of frozen image and image-text embeddings across a broad set of downstream formats. It does not measure end-to-end generative performance, instruction following in full multimodal dialogue, or performance after task-specific fine-tuning. Readers should therefore interpret strong MIEB performance as evidence of broad embedding utility, not as evidence that a model solves multimodal reasoning in general.

The benchmark also has important limitations. Because MIEB standardizes existing public datasets, it inherits their annotation artifacts, domain imbalances, and potential contamination risks. Some categories contain more tasks than others, so aggregate means should always be read alongside the category breakdown. The linear probing setting introduces small supervised adaptation, which is useful diagnostically but differs from strict zero-shot evaluation. Finally, multilingual evaluation is still concentrated in retrieval and Visual STS rather than all eight categories.

These limitations define the benchmark's scope. MIEB is useful because it separates what current embeddings already do well from what still fails under broader evaluation.

8. Conclusion

We presented MIEB, a unified benchmark for image and image-text embeddings spanning 130 tasks, 8 categories, and 38 languages. By standardizing evaluation across retrieval, document understanding, classification, clustering, compositionality, vision-centric QA, and Visual STS, MIEB exposes trade-offs that narrow benchmarks hide. Our experiments show that no single model dominates across all categories: contrastive encoders remain strongest on classification-oriented tasks, while MLLM-based embeddings are stronger on multilingual retrieval and visual text understanding. MIEB-lite further shows that much of this diagnostic value can be retained at much lower evaluation cost.

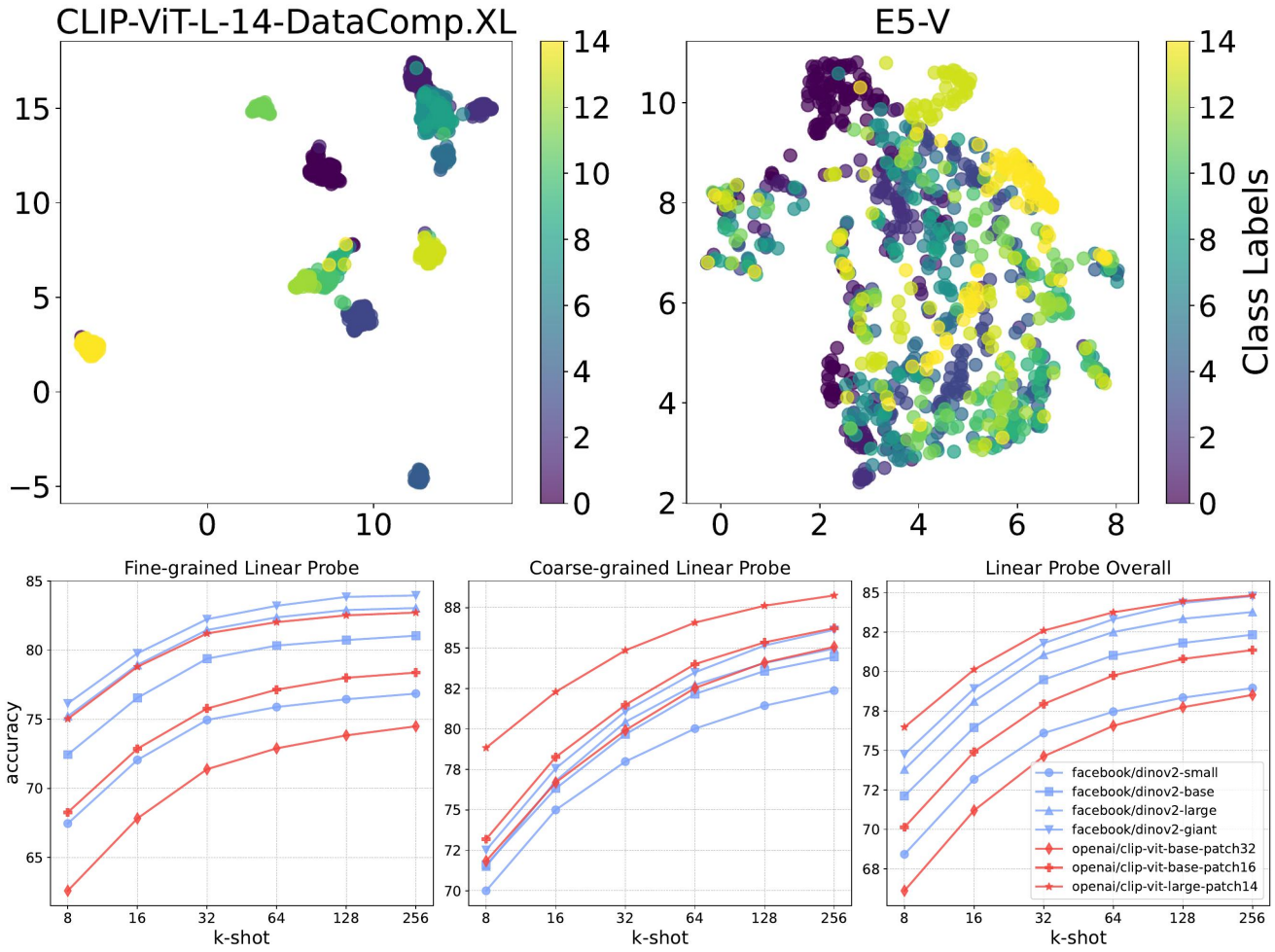

Figure 2. **Qualitative diagnostic slices.** Top: a Dog15 clustering slice where CLIP-ViT-L-14-DataComp.XL separates classes more cleanly than E5-V. Bottom: linear probing improves quickly with more shots, then begins to flatten.

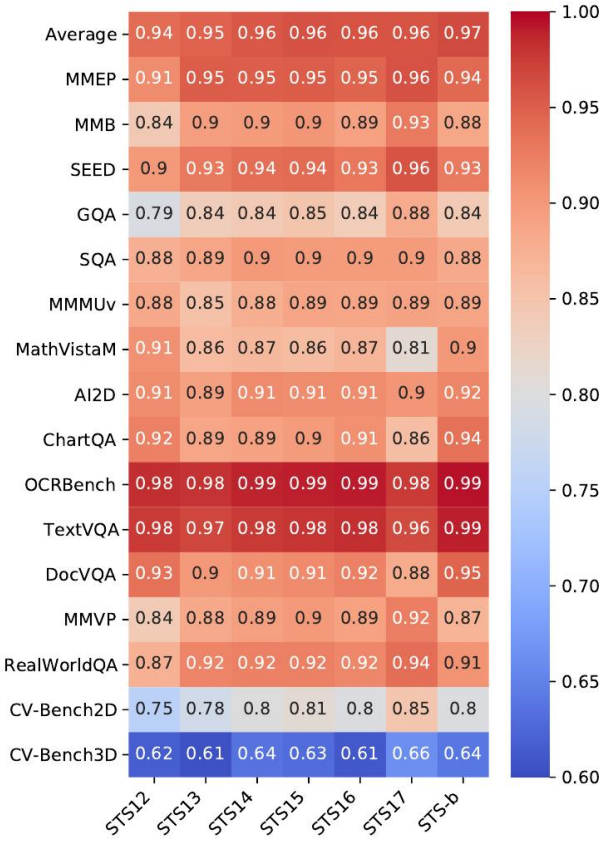

Figure 3. **Visual STS vs. MLLM benchmarks.** Correlation is strongest for OCR-heavy tasks and weaker for 3D-heavy perception.

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